WHITE PAPER · MARCH 2026

Anatomy of a $53 Billion Mega-Deal

A quantitative valuation framework dissecting Chevron's acquisition of Hess Corporation through six investment banking methodologies ; built entirely in Python.

REPOSITORY	github.com/DogInfantry/hess-chevron-valuation-analysis
STACK	Python · NumPy · Pandas · Matplotlib · SciPy · yfinance · FRED API
MODELS	DCF · NAV · LBO · Trading Comps · Precedent Transactions · Accretion/Dilution

For Educational & Portfolio Demonstration Purposes

Contents

Executive Summary 3
Transaction Overview 3
Strategic Rationale 3
Hess Corporation: Target Profile 4
Stabroek Block, Guyana 4
Financial Summary (FY2023 / FY2022) 4
Production Growth (KBOEPD) 5
Precedent Transactions Analysis 5
DCF & Net Asset Value Analysis 6
5.1 WACC Computation 6
5.2 Five-Year Projection (2024–2028) 6
5.3 DCF Sensitivity Matrix 6
5.4 Net Asset Value Build 7
LBO Floor Valuation 8
Maximum Affordable Price by Target IRR 8
Accretion/Dilution & Synergy Analysis 8
Standalone vs. Pro Forma 8
Synergy Sensitivity 9
Valuation Football Field 9
Deal Verdict 10
The Structural Bifurcation 10
For Hess Shareholders 10
For Chevron Shareholders 10

S E C T I O N 0 1

Executive Summary

On October 23, 2023, Chevron Corporation announced an all-stock acquisition of Hess Corporation valued at $53 billion in equity ($60 billion enterprise value). This white paper presents a six-methodology valuation framework, built entirely in Python that evaluates the strategic and financial merits of the transaction. The analysis reveals a structural bifurcation: cash-flow-based models (DCF, LBO) produce valuations materially below Chevron's $171/share offer, while asset-based and market-based approaches (NAV, Trading Comps, Precedent Transactions) generate values more consistent with the deal price.

This gap quantifies the "Guyana premium",the option value embedded in Hess's 30% interest in the Stabroek Block, an 11-billion-barrel deepwater resource that a five-year DCF horizon cannot capture.

Equity Value	Enterprise Value	Deal EV/EBITDA	Guyana Resource	Y1 EPS Impact
$53B	**$60B**	**11.6x**	**11B boe**	**−6.9%**

Deal closed July 18, 2025 after ICC arbitration ruled in Chevron's favour against ExxonMobil's preemptive rights claim. Post-close combined production: ~3.83M boe/d.

S E C T I O N 0 2

Transaction Overview

Parameter	Detail
Acquirer / Target	Chevron Corporation (CVX) / Hess Corporation (HES)
Consideration	All-stock merger (1.025 CVX shares per HES share)
Implied Price	$171/share (10.3% premium to $155.39 unaffected)
Equity / Enterprise Value	$53B / $60B (incl. $6.6B net debt)
Deal EV/EBITDA	11.6x FY2023 EBITDA ($5,176M)
Announced / Closed	October 23, 2023 / July 18, 2025
HES Shareholder Ownership	~14.7% of combined entity
CVX Advisors	Morgan Stanley, Evercore
HES Advisors	Goldman Sachs, J.P. Morgan
Key Regulatory Hurdle	ExxonMobil ICC arbitration (resolved Jul 2025 in CVX favour)

Strategic Rationale

1. **Resource Depth via Guyana.** 30% interest in the Stabroek Block: 11B+ boe discovered, $35/bbl breakeven, 6+ FPSOs planned by 2030. Analysts estimate Guyana represents ~70% of Hess's standalone valuation ($40–46B net to Hess).

2. **Portfolio Diversification.** Adds 465,000 net Bakken acres, GoM deepwater positions, and SE Asian gas assets—broadening Chevron's portfolio to 2.5M+ net onshore acres across three premier U.S. shale basins.
3. **Cash Flow Accretion.** Chevron projects doubling of free cash flow by 2027 and production growth extending into the 2030s. post-close combined production: ~3.83M boe/d.
4. **Consolidation Positioning.** Cements Chevron as the #2 global integrated major by production, maintaining strategic parity with ExxonMobil in an era where scale determines capital efficiency.

S E C T I O N 0 3

Hess Corporation: Target Profile

Hess Corporation, a pure-play upstream E&P headquartered in New York, produced 391,000 boe/d in FY2023 across four core regions, generating $10.5B in revenue and $5.2B in EBITDA. Production grew 43% from 2019 to 2023, driven almost entirely by the Guyana Stabroek Block ramp.

Stabroek Block, Guyana

Metric	Value
Block Size	6.6 million acres
Operator / Partners	ExxonMobil (45%) / Hess (30%) / CNOOC (25%)
Discovered Resources	~11 billion BOE recoverable
Operating FPSOs (2024)	3 (Liza-1, Liza-2, Payara)
Planned FPSOs by 2030	6+ (with debottlenecking upside of 12–20%)
Gross Production (2024E)	~640 KBOEPD
Breakeven Cost	~$35/bbl Brent
Scope 1+2 Emissions Intensity	Among lowest in global upstream

Financial Summary (FY2023 / FY2022)

Metric ($M)	FY2023	FY2022
Revenue	$10,482	$11,314
Production Costs	$4,521 (43.1%)	$4,480 (39.6%)
EBITDA	$5,176	$6,039
DD&A	$2,012	$1,945
EBIT	$3,164	$4,094
Net Income	$2,103	$2,095
Capital Expenditures	$3,620	$2,996
Total Debt / Cash	$8,268 / $1,715	$8,351 / $1,880
Production (KBOEPD)	391	356

Source: Hess Corporation 10-K filings, FY2022 and FY2023. HES delisted post-close; all data hardcoded from SEC filings.

Production Growth (KBOEPD)

2019	2020	2021	2022	2023
273	321	315	356	**391**

+43% cumulative growth (2019→2023). Nearly all attributable to Guyana Stabroek Block FPSO ramp-up.

S E C T I O N 0 4

Precedent Transactions Analysis

Eleven E&P M&A transactions from 2020–2024 form the precedent universe, the largest consolidation wave since the early 2000s, totaling over $200 billion in announced deal value.

Acquirer	Target	Date	EV ($B)	EV/EBITDA	Premium
ExxonMobil	Pioneer NR	Oct 23	64.5	6.2x	18%
Chevron	Hess Corp.	Oct 23	60.0	**8.4x**	10%
Diamondback	Endeavor	Feb 24	26.0	7.1x	–
ConocoPhillips	Marathon Oil	May 24	22.5	5.8x	15%
ConocoPhillips	Concho Res.	Oct 20	13.3	5.9x	15%
OXY	Crown Rock	Dec 23	12.0	5.5x	–
Chevron	PDC Energy	May 23	7.6	4.8x	11%
Chesapeake	Southwestern	Jan 24	7.4	5.2x	12%
Civitas	Tap Rock	Jun 23	4.7	4.5x	–
APA Corp	Callon Petro.	Jan 24	4.5	4.1x	8%
Devon	Validus	Jul 23	1.8	4.0x	–
Summary				**Med: 5.5x**	**Med: 12%**

Key Insight

Chevron's 8.4x EV/EBITDA is 53% above the sector median (5.5x), the highest in the sample. The premium quantifies the market's view that Guyana's resource quality is unmatched. Yet Chevron's 10.3% 1-day premium is below the deal median of ~12%, suggesting effective acquirer negotiation.

SECTION 05

DCF & Net Asset Value Analysis

The DCF and NAV analyses together expose the central tension of this transaction: the gap between what a cash-flow model says Hess is worth and what a resource-based model implies.

5.1 WACC Computation

Input	Value	Source
Risk-Free Rate (Rf)	4.30%	U.S. 10-Year Treasury
Unlevered Beta (βu)	1.10	Damodaran E&P sector
Equity Risk Premium (ERP)	5.50%	Damodaran implied ERP
Cost of Equity (Ke = Rf + β × ERP)	**10.35%**	CAPM
Pre-Tax Cost of Debt (Kd)	4.50%	Hess blended rate
Tax Rate	21.0%	U.S. statutory
After-Tax Cost of Debt	3.56%	Kd × (1 – t)
Equity Weight / Debt Weight	85.2% / 14.8%	At unaffected price
→ WACC	**9.33%**	Weighted average

5.2 Five-Year Projection (2024–2028)

Revenue built on asset-level production forecasts: Guyana FPSO ramp-up at 10% declining to 6%, with FY2023 cost margins as the base. UFCF derived via EBIT × (1–t) + DD&A – CapEx.

Metric	2024E	2025E	2026E	2027E	2028E
Prod. Growth	10%	9%	8%	7%	6%
Revenue ($M)	11,530	12,568	13,573	14,523	15,395
EBITDA ($M)	5,694	6,206	6,703	7,172	7,603
DD&A ($M)	2,213	2,413	2,606	2,788	2,956
NOPAT ($M)	2,750	2,996	3,237	3,463	3,671
CapEx ($M)	3,982	4,340	4,688	5,016	5,316
UFCF ($M)	**981**	**1,069**	**1,155**	**1,235**	**1,311**

5.3 DCF Sensitivity Matrix

Implied share price at varying WACC (columns) and terminal growth rate (rows). Base case highlighted.

TG \ WACC	7.5%	8.5%	9.5%	10.5%	11.5%
1.0%	$68	$40	$19	$3	($10)

1.5%	$85	$51	$27	$9	($5)
2.0%	$108	$66	**$37**	$15	$0
2.5%	$140	$87	$50	$24	$6
3.0%	$190	$118	$70	$37	$14

The DCF Problem

At base-case WACC (9.5%) and 2% terminal growth, the DCF implies ~$37/share; far below Chevron's $171 offer. This is not a modelling error. A 5-year horizon structurally cannot capture the option value of 11B barrels of low-cost reserves with a multi-decade development runway. Terminal value represents ~63% of enterprise value. The DCF provides a conservative floor; the NAV captures what the market actually prices.

5.4 Net Asset Value Build

The NAV values Hess's reserves on a per-barrel basis across six discrete asset segments, capturing the full resource base that the DCF cannot.

Asset	MMBOE	$/BOE	Value ($M)	Category
Guyana — Proved Developed	350	$15.0	**$5,250**	Proved Developed
Guyana — Proved Undeveloped	550	$10.0	**$5,500**	Proved Undeveloped
Guyana — Probable/Contingent	1,800	$5.0	**$9,000**	Probable/Contingent
Bakken (N. Dakota)	450	$10.0	**$4,500**	Proved Developed
Gulf of Mexico	120	$12.0	**$1,440**	Proved Developed
Southeast Asia (Gas)	200	$6.0	**$1,200**	Proved Developed
Gross Asset Value	3,470		**$26,890**	
Less: Net Debt			($6,553)	
NAV per Share			**$66**	(306.2M shares)

Guyana represents 73% of gross NAV ($19,750M of $26,890M). The DCF-to-NAV gap quantifies the option value of undeveloped reserves. This is precisely why strategic acquirers pay premiums that financial models anchored to 5-year cash flows cannot justify. In a fairness opinion, the DCF provides a conservative floor; the NAV captures what the market actually prices.

SECTION 06

LBO Floor Valuation

Caveat: *A $60B LBO exceeds the capacity of any single financial sponsor. This model demonstrates methodology and establishes a valuation floor,it does not represent a feasible transaction.*

Entry EV	Total Debt	Sponsor Equity
$60,000M	**$20,704M**	**$39,296M**
11.6x EBITDA	4.0x EBITDA	65.5% of EV

Maximum Affordable Price by Target IRR

Target IRR	Req. MOIC	Max Entry EV	Max $/Share	vs. Offer
25%	3.05x	~$30B	**~$55**	−68%
20%	2.49x	~$38B	**~$85**	−50%
15%	2.01x	~$48B	**~$120**	−30%

LBO Verdict

Even at a 15% IRR threshold (below standard PE hurdle rates), the maximum affordable price is ~$120/share,30% below Chevron's offer. No financial sponsor can match the strategic premium. The $51 gap between $120 and $171 is the quantified value of being a strategic acquirer with operational synergies and a multi-decade investment horizon.

SECTION 07

Accretion/Dilution & Synergy Analysis

For an all-stock deal, the critical question for CVX shareholders: does this increase or decrease my earnings per share?

Standalone vs. Pro Forma

Metric	Value
CVX Standalone EPS (FY2023)	$13.17
CVX Net Income	$25,400M
CVX Shares Outstanding	1,930M
New Shares Issued (306.2M × 1.025)	314M
Combined Shares	2,244M
HES Net Income	$2,103M

Combined Net Income (no synergies)	$27,503M
Pro Forma EPS (no synergies)	**$12.26**
Accretion / (Dilution)	**(6.9%) — DILUTIVE**

Synergy Sensitivity

Pre-Tax Syn.	After-Tax	Comb. NI	PF EPS	Acc/(Dil)
$0	$0	$27,503M	**$12.26**	-6.9%
$500M	$395M	$27,898M	**$12.44**	-5.5%
$1,000M	$790M	$28,293M	**$12.61**	-4.3%
$1,500M	$1,185M	$28,688M	**$12.79**	-2.9%
$2,593M	$2,048M	$29,551M	**$13.17**	0.0%

The Real Thesis

Even at Chevron's guided $1B synergies, the deal remains ~4.3% dilutive. Breakeven requires $2.6B pre-tax,well above typical E&P combination targets. Chevron is not buying Hess for today's earnings. It is buying Guyana's future production ramp. Short-term EPS dilution is the price of decades of low-cost growth.

S E C T I O N 0 8

Valuation Football Field

The signature deliverable: seven methodologies, one synthesis. All implied share prices versus Chevron's $171 offer.

Methodology	Low	Midpoint	High	vs. $171 Offer
Premium-Implied (8–18%)	$168	**$174**	$183	*Brackets offer*
NAV ($60–$80/bbl)	$130	**$168**	$210	*Near offer*
Trading Comps	$92	**$145**	$178	*Range captures offer*
Precedent Transactions	$72	**$115**	$155	*Below offer*
52-Week Trading Range	$120	**$140**	$160	*Below offer*
LBO Floor (15–25% IRR)	$55	**$85**	$120	*Well below*
DCF (MC P10–P90)	$15	**$37**	$60	*Far below*

SECTION 09

Deal Verdict

The Structural Bifurcation

Cash-flow models (DCF, LBO) say Hess is worth far less than $171. Resource and market models (NAV, Comps, Precedent Premiums) say it is worth about that. The gap is not a modelling error,it is the Guyana premium: the option value on 11 billion barrels that a 5-year cash flow horizon structurally cannot capture.

For Hess Shareholders

- 10.3% premium sits at the low end of E&P deal premiums (sector median: ~12%).
- All-stock structure preserves tax deferral and ongoing Guyana upside through CVX equity.
- Premium-implied range ($168–$183) brackets the $171 offer,market-consistent pricing.
- Arbitration resolution removed the key overhang that had compressed the stock.

For Chevron Shareholders

- 6.9% EPS dilutive in Year 1; breakeven requires $2.6B synergies (above guided $1B).
- Post-close: 3.83M boe/d, #2 global integrated major by production.
- Guyana extends production growth into the 2030s at breakeven costs of ~$35/bbl.
- $2.3B open-market HES purchase during arbitration signals high management conviction.

Bottom Line

Chevron's $171 offer is defensible on a resource basis (NAV) and consistent with market-implied pricing (premium analysis), but requires acceptance of near-term earnings dilution as the cost of acquiring a generational asset. The deal is fair but not generous,Chevron negotiated effectively, and Hess shareholders retain upside through CVX equity. The true test is whether Guyana delivers on its multi-decade production promise.

Disclaimer

This analysis is produced for educational and portfolio demonstration purposes only. It does not constitute investment advice, a solicitation to buy or sell securities, or a recommendation regarding the Chevron–Hess transaction. All financial data and projections are based on publicly available information from SEC filings, press releases, and market data providers. The author has no material position in any securities discussed.

github.com/DogInfantry/hess-chevron-valuation-analysis · Built in Python · March 2026